Homework 1: Image Enhancement with Classical Methods

Computer Vision, Fall 2026 | Melat Habtemariyam Ekubay, VR545975

1. Objective

The goal is to build and compare a small pipeline that enhances colour images affected by low contrast, non-uniform illumination and noise, using only classical image processing (OpenCV and NumPy; no neural networks). The aim is not to reconstruct the original scene perfectly, but to justify the choices, observe the effect of each method and discuss the trade-offs between detail visibility, noise and natural appearance. **Important:** the original photographs are used as *references* only because the degradations are synthetic. In real-world scenarios a clean reference is usually not available, so PSNR and SSIM could not be computed and the evaluation would rely on visual judgement.

2. Images, sources and licenses

Four real photographs were used (Table 1). Licenses are those stated in the scikit-image documentation; the images are included in the repository (`data/original/`), together with `data/SOURCES_AND_LICENSES.md`. For each image a degraded version was generated (Section 3) and both versions are kept.

Image	Content and challenge	Source	License
astronaut	Portrait; dark suit and helmet next to bright background (512x512)	NASA Great Images database, via scikit-image	Public domain
chelsea	Cat close-up; fine fur texture, whiskers, soft mid-tones (300x451)	Photo: S. van der Walt, via scikit-image	CC0
coffee	Espresso cup; dark shadow areas and wood grain (400x600)	Photo: R. Michetti (Pikolo Espresso Bar), via scikit-image	CC0
rocket	Launch pad at dusk; strongly non-uniform illumination, dark sky (427x640)	SpaceX Photos (Flickr), via scikit-image	Public domain (released by SpaceX)

Table 1. Image set.

3. Methods and parameters

Degradation (fixed for all methods). Contrast reduction *out = a(in - 128) + 128* with a = 0.45, followed by additive Gaussian noise with sigma = 5 grey levels (0-255 scale), clipping to [0, 255] and rounding to 8 bits. The random generator uses the fixed seed 42.

Colour handling. Methods 2-4 convert the BGR image to CIE-Lab, modify only the luminance channel L and keep the colour channels a, b unchanged, so the enhancement itself does not mix colours.

M1 Baseline: the degraded image, unprocessed. **M2 Linear contrast stretching:** on L, the 1st and 99th percentiles are mapped to 0 and 255 (percentile clipping avoids control by a few extreme pixels). A supplementary variant stretches each B, G, R channel separately, only to discuss colour shifts. **M3 Histogram equalization:** global equalization (`cv2.equalizeHist`) of L. **M4 Gamma correction:** on L, *out = 255(in/255)^gamma*, where gamma is estimated from the degraded image alone so that its mean luminance becomes 0.5 (resulting gamma: astronaut 1.03, chelsea 1.05, coffee 0.97, rocket 0.79).

Metrics (always against the original): PSNR (own NumPy implementation, cross-checked with `cv2.PSNR`), SSIM (scikit-image, colour, data range 255) and mean absolute error (MAE, in the CSV). We also report the luminance standard deviation (contrast) and the mean Lab a/b difference to the original (colour shift).

4. Results

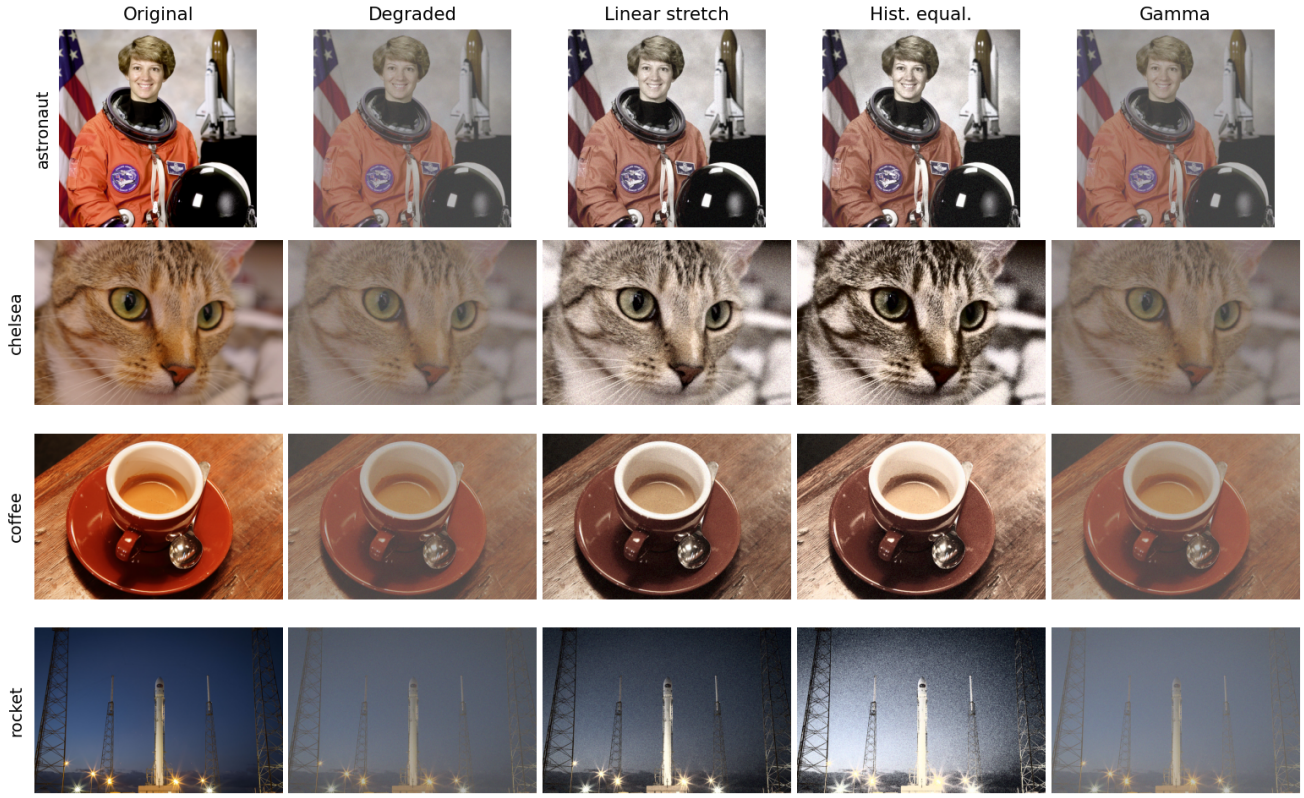


Figure 1. Original, degraded and enhanced results for the four images (same size and scale for all panels). Larger per-image figures with zoom and histograms are in `results/figures/`.

Image	M1 Baseline	M2 Stretch (L)	M3 Hist. eq. (L)	M4 Gamma (L)	Suppl.: stretch per channel
astronaut	14.96 / 0.569	**21.46 / 0.630**	20.13 / 0.579	15.00 / 0.569	24.32 / 0.579
chelsea	20.25 / 0.696	16.08 / 0.573	14.70 / 0.426	**20.60 / 0.701**	16.57 / 0.486
coffee	15.24 / 0.577	**18.76 / 0.639**	15.63 / 0.546	15.02 / 0.571	23.70 / 0.569
rocket	16.14 / 0.607	**22.31 / 0.475**	10.11 / 0.197	12.72 / 0.545	16.66 / 0.349
Mean	16.65 / 0.612	19.65 / 0.579	15.14 / 0.437	15.83 / 0.596	20.31 / 0.496

Table 2. PSNR (dB) / SSIM with respect to the original; bold = highest PSNR among M1-M4. All values are computed by the code (`results/metrics.csv`, which also contains MAE).

5. Discussion

Overall. Linear stretching of the luminance gives the highest PSNR on three of the four images and the best mean PSNR (19.65 dB versus 16.65 dB for the baseline). Histogram equalization has the lowest mean scores (15.14 dB, SSIM 0.437). Gamma correction is almost the identity because the contrast reduction pulls the mean luminance towards the middle grey: the estimated gamma is close to 1 for three images, so it cannot fix low contrast. It only changes the result clearly for the dark *rocket* image (gamma 0.79).

Case 1: rocket (non-uniform illumination). The gantry towers and the rocket body become much easier to see after linear stretching (luminance contrast 14.8 -> 43.8; the original has 33.2), and PSNR rises from 16.14 to 22.31 dB. However SSIM *decreases* (0.607 -> 0.475), because the noise in the large flat sky is amplified and the blue dusk sky turns grey: the degradation compressed the colour channels too, and luminance-only stretching does not restore them. Histogram equalization performs worst here (10.11 dB, SSIM 0.197): the sky covers most pixels, so its grey levels are spread over half of the range, producing a bright, strongly noisy sky and saturated highlights. Gamma (brightening) lowers PSNR to 12.72 dB because the original dusk scene is genuinely dark. **Preferred method: linear stretch**, as it reveals structure with the most natural result, but the amplified noise is clearly visible (Figure 2).

Case 2: chelsea (numbers and visual quality disagree). Visually, stretching and equalization make the fur texture, whiskers and the structure of the eye much clearer than the washed-out degraded image. Yet both have *lower* PSNR/SSIM than the unprocessed baseline (16.08 and 14.70 dB versus 20.25 dB). The reason is that the original is itself a low-contrast photograph (luminance std 32.7); the methods push it to 52.5 (stretch) and 73.7 (equalization), going *beyond* the reference, and they amplify noise, and both effects count as error. Conversely, gamma has the highest numbers on this image (PSNR 20.60 dB, SSIM 0.701) only because gamma = 1.05 leaves the image almost unchanged and flat. So a numerical improvement did not mean a more pleasing image, and a more pleasing image did not mean a better score. **Preferred method: linear stretch** (equalization looks harsh and noisy in the fur and background).

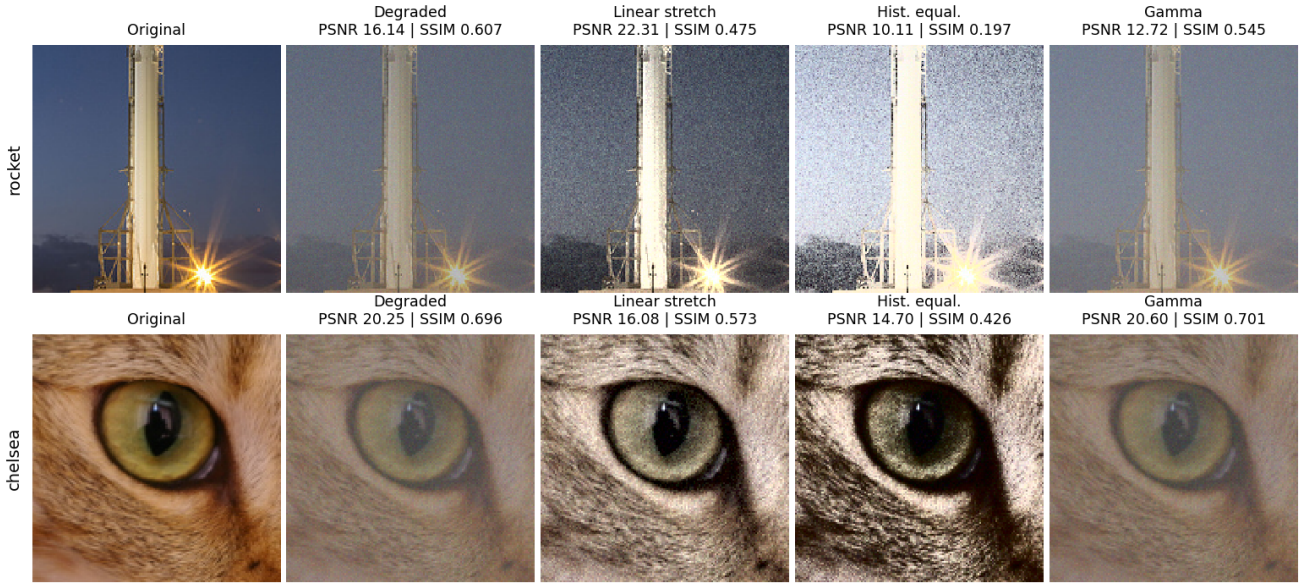


Figure 2. Zoomed regions (boxes marked in `results/figures/*_comparison.png`) for the two analysed cases, *rocket* and *chelsea*. Titles give PSNR (dB) | SSIM.

Noise, colours and detail. Every method that expands contrast expands noise by a similar factor (for linear stretching the luminance contrast is multiplied by 2.2-3.5, depending on the image, and the noise is amplified accordingly), and equalization does this most in flat regions. Because M2-M4 leave the a/b channels untouched, hues are preserved but the colours stay as desaturated as in the degraded input (colour shift equal to the baseline, e.g. astronaut 9.33 vs 9.32). The supplementary per-channel stretch restores saturation on *astronaut* and *coffee* (highest PSNR there: 24.32 and 23.70 dB) but introduces a strong blue cast on *rocket* (colour shift 7.15 -> 16.39), so luminance-only processing is safer. Percentile clipping and equalization saturate some highlights (cup rim, launch lights, sky), losing detail there; none of the methods can recover detail already hidden by noise, and none removes noise.

6. Limitations and trade-offs

(i) Only four public photographs were used and they were not captured under extreme lighting; low contrast is mostly introduced synthetically. (ii) The degradation is linear, so linear stretching is almost its exact inverse, which favours M2 in the numbers; real shadows are not removed by a global linear mapping. (iii) Single noise seed and single parameter set, so no confidence intervals. (iv) Global equalization was used; local methods (CLAHE) or denoising combined with stretching (e.g. a bilateral filter) could reduce the noise amplification but were not part of this study. (v) PSNR and SSIM are global measures and do not fully capture perceived quality, so no method was chosen on a single metric alone.

7. Repository and reproducibility

Repository: **https://github.com/melu7/hw1-image-enhancement**. It contains the Python code, original and degraded images, processed images, `results/metrics.csv` and a README with the Python version (3.12; 3.9+ supported), dependencies and the command `python run_experiment.py`, which runs the whole pipeline from the original images with fixed seed 42 and produces byte-identical outputs on repeated runs.

References. G. Bradski, The OpenCV Library, *Dr. Dobb's Journal*, 2000. S. van der Walt et al., scikit-image: image processing in Python, *PeerJ* 2:e453, 2014. Z. Wang et al., Image quality assessment: from error visibility to structural similarity, *IEEE Trans. Image Process.* 13(4):600-612, 2004.